
Childcare Before Age Three and Long-Term Academic Achievement

Evidence from a Lottery in Barcelona

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Motivation

- Subsidizing early (0–3) childcare is an increasingly popular family policy. Two main goals:
 - (1) support parental labour supply;
 - (2) promote child development

Motivation

- Subsidizing early (0–3) childcare is an increasingly popular family policy. Two main goals:
 - (1) support parental labour supply;
 - (2) promote child development
- The first three years are a **critical period**: brain development is particularly rapid and malleable, laying the foundation for “skill begets skill” [Cunha and Heckman, 2007]
- What do we know about the effects? [Duncan et al., 2023]
 - For **older preschoolers** (age 3+): consistently positive long-run effects for disadvantaged children
 - For **younger children** (ages 0–2): evidence is mixed, and almost entirely short- or medium-run
- Theory is ambiguous: what matters is the **quality of childcare relative to the counterfactual** — the care it displaces — which varies with family background and is hard to pin down empirically

This Paper

- We provide **causal evidence** on the effect of public childcare (ages 0–3) on **long-run** academic achievement
- We exploit lottery-based admission to public childcare in Barcelona (2005–2010)
- Linked administrative data: lottery records matched to **standardized test scores at age 12**
- **Finding:** Admitted children score **0.08 SD lower** at age 12
- Our reading: for this population, public childcare was of **lower quality than the care it displaced**

Contributions in detail →

Public Childcare in Barcelona

- Barcelona operates **public nurseries** (*escoles bressol municipals*) for ages 0–3
 - Subsidized (not free), limited slots
 - There are also (more expensive) private nurseries
- High quality reputation; **excess demand**
- **Centralized allocation:** Boston-style mechanism with lottery tie-breaking
 1. Families with priority points (siblings, proximity, low income)
 2. 76% of applicants have identical priority score → lottery
- Families may rank **strategically**, so the “first choice” is a strategic one. This does not threaten **internal** validity (the lottery number is random within a risk set), but it matters for interpretation

Characteristics → Strategic applications → If rejected →

Empirical Specification

$$y_{it} = \gamma \cdot \text{Admitted}_{it} + D'_{it}\alpha + X'_{it}\beta + \varepsilon_{it}$$

- y_{it} : test score at age 12
- Admitted_{it} : offered seat at first-choice center
- D_{it} : **lottery fixed effects** (center $\times$ year $\times$ level)
- X_{it} : child and family controls
- Standard errors clustered at the lottery level

Identification: Within each lottery, compare winners vs. losers who applied to the **same center, same year, same level**, with the **same priority score**. Only difference: random lottery number.

We have **540 such lotteries** with 7,980 children. We estimate **ITT** (intent-to-treat).

Winners and losers are **balanced** on all observables, and there is **no differential attrition**.

Balance $\rightarrow$ Attrition $\rightarrow$ ITT interpretation $\rightarrow$

IDESCAT-linked individual-level data from four sources:

- **Childcare Applications (IMEB, 2005–2010)**
 - Preference rankings, priority points, lottery outcomes
 - Family characteristics at application
- **School Records (Departament d'Educació)**
 - Enrollment, primary school attended
- **Test Scores (Consell Superior d'Avaluació)**
 - *Avaluació de sisè de primària*: Catalan, Spanish, English, Math
- **Municipal Census (Padró Municipal)**
 - Household composition, parental demographics

Test details → Supplementary data → Sample construction →

Main Result: Average Test Scores

	(1) Average Test Score	(2) Average Test Score	(3) Average Test Score	(4) Average Test Score	(5) Average Test Score
Admitted	-0.157*** (0.018)	-0.136*** (0.027)	-0.085*** (0.028)	-0.085*** (0.028)	-0.082*** (0.026)
Observations	16557	7980	7980	7980	7980
FE Prob.			Yes		
FE Lottery				Yes	Yes
Controls					Yes
Sample	All	Lotteries	Lotteries	Lotteries	Lotteries

Headline: Admission to public childcare **reduces** test scores at age 12 by **0.08 SD**

cf. Fort et al. [2020]: -0.05 SD/month in Bologna (very affluent sample; our effect ≈ -0.01 SD/month)

How big is 0.08 SD? → Distribution → Results T_all →

Results by Subject

	(1) Average Test Score	(2) Test Score Catalan	(3) Test Score Spanish	(4) Test Score English	(5) Test Score Maths
Admitted	-0.082*** (0.026)	-0.059** (0.026)	-0.086*** (0.027)	-0.017 (0.027)	-0.098*** (0.027)
Observations	7980	7980	7980	7980	7980

- Negative effects on Catalan, Spanish and Math; **no effect on English**
- English is a **natural placebo**: (most) children under 3 in Barcelona are not exposed to English — neither at home nor in the nursery — so it is acquired only through schooling from age 3
- $\Rightarrow$ Consistent with effects operating through **early language exposure**

English placebo in detail $\rightarrow$

Results by Age at Entry

	(1) Average Test Score	(2) Test Score Catalan	(3) Test Score Spanish	(4) Test Score English	(5) Test Score Maths
<i>Panel A: Age 0–1 (entry at 4–12 months)</i>					
Admitted	-0.125* (0.073)	-0.161** (0.076)	-0.175** (0.087)	-0.021 (0.077)	-0.041 (0.079)
Observations	792	792	792	792	792
Mean	0.21	0.11	0.09	0.19	0.25
<i>Panel B: Age 1–2 (entry at 12–24 months)</i>					
Admitted	-0.074* (0.040)	-0.076** (0.039)	-0.081* (0.041)	-0.001 (0.038)	-0.078* (0.040)
Observations	4336	4336	4336	4336	4336
Mean	0.02	0.03	0.00	0.03	0.00
<i>Panel C: Age 2–3 (entry at 24–36 months)</i>					
Admitted	-0.087** (0.041)	-0.018 (0.040)	-0.074* (0.040)	-0.044 (0.045)	-0.135*** (0.043)
Observations	2852	2852	2852	2852	2852
Mean	-0.09	-0.08	-0.03	-0.10	-0.07

Language penalties largest at **youngest** entry; Math penalties at **oldest**

Heterogeneity: Parental Education

	(1)	(2)	(3)	(4)
	Average Test Score	Average Test Score	Average Test Score	Average Test Score
Admitted x High Edu	-0.090*** (0.033)	-0.189** (0.073)	-0.075 (0.048)	-0.084 (0.057)
Admitted x Low Edu	-0.074* (0.038)	-0.014 (0.123)	-0.072 (0.059)	-0.091 (0.056)
Observations	7980	792	4336	2852
Mean	-0.00	0.21	0.02	-0.09
Sample	All levels	Age 0–1	Age 1–2	Age 2–3
p-value (H0: equal)	0.75	0.17	0.96	0.93

- Negative effects for **both** high- and low-education families; the difference is small and not significant
- Different from Almås et al. [2026]: in Norway the **positive** effects come entirely from children of mothers without a high-school diploma (+0.28 SD in grade-9 maths).

Sample selection → Who applies? → Educ. var → Income → Gender →

Why Negative? Quality Relative to the Counterfactual

The sign of the effect depends on the quality of public childcare **relative to the care it displaces**. Two questions:

1. What care do rejected families use instead?

- Administrative enrollment data + a survey of families with young children that identifies rejected applicants

2. How good is Barcelona's public childcare?

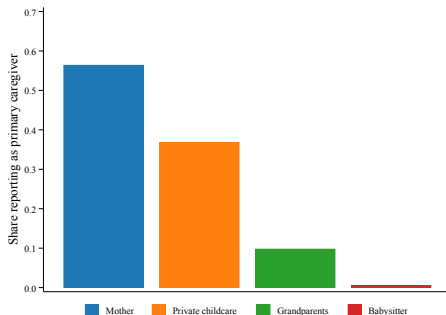
- Child-teacher ratios, benchmarked against settings with positive effects

Not a later-schooling artifact: admission shifts children towards public primary schools, but adding **school fixed effects** leaves the estimates unchanged, and there is no effect on peer test scores.

School sorting → Peer quality →

What Do Rejected Families Do?

We do not observe our control group's care. A survey of Barcelona families with young children identifies rejected applicants (IGOP 2020, *not* our sample):



- A **mix**: mostly parental care, but **private childcare for over a third**
- Matches admin data: at ages 0–1, ~89% are in **no formal childcare**; private enrollment rises only at ages 2–3

Is It a Public vs. Private Story?

	(1) Average Test Score	(2) Average Test Score	(3) Average Test Score	(4) Average Test Score
Admitted $\times$ High Private Availability	-0.066* (0.037)	-0.086 (0.082)	-0.096 (0.058)	-0.070 (0.049)
Admitted $\times$ Low Private Availability	-0.088** (0.037)	-0.204 (0.130)	-0.051 (0.051)	-0.118** (0.049)
Observations	7772	782	4433	4115
Mean	0.00	0.20	0.03	-0.06
Sample	All levels	Age 0–1	Age 1–2	Age 2–3
p-value (H0: equal)	0.66	0.43	0.55	0.49

- Negative effects in **both** high- and low-private-availability districts
- $\Rightarrow$ Not **only** a public-vs-private story. But private care *is* part of the counterfactual.

Variable def. $\rightarrow$

Quality: Child–Teacher Ratios

Barcelona's public childcare operates at **maximum permitted ratios**:

Setting	Age 0–1	Age 1–2	Age 2–3
Barcelona (regulatory max.)	8:1	13:1	20:1
Norway [Almås et al., 2026]	3:1 (national guideline, ages 1–2)		
Bologna, Italy [Fort et al., 2020]	4:1	6:1	8:1
Perry Preschool*		3:1 to 6:1	
Abecedarian*		3:1 to 6:1	

Barcelona's ratios are **2–7 times worse** than the settings that find positive effects.

Quality: Child–Teacher Ratios



Conclusion

Access to public childcare before age 3 **reduced** test scores at age 12 by **0.08 SD**.

Interpretation — quality relative to the counterfactual:

Barcelona's public nurseries run at the regulatory maximum (8:1–20:1) and displace a mix of parental, grandparental and private care that is plausibly of higher quality.

Not a story about disadvantaged children: effects are similar across parental education and neighbourhood income — unlike Norway [Almås et al., 2026] and Italy [Fort et al., 2020].

Policy: the message is **not** that early childcare is harmful, but that policy has prioritized **expansion** over **quality**.

Work in progress: a new survey of applicants will let us measure **non-cognitive outcomes**, the **counterfactual mode of care** directly, and **parental labour-market outcomes**.

Questions?

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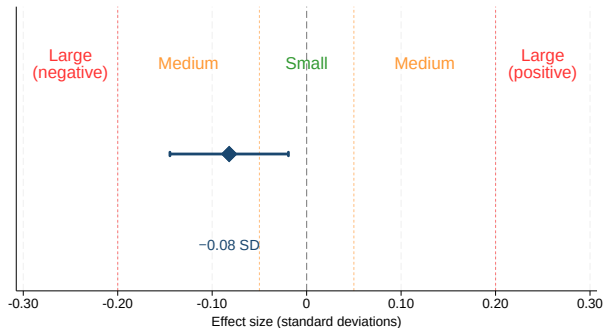
Our Contribution

We provide **causal evidence** on childcare before age 3:

1. **Representative sample** — Not restricted to very high or low income families (cf. Fort et al., 2020)
2. **Standardized national exams** — Test scores at age 12: the **longest horizon** in the 0–2 lottery literature
3. **Credible randomization** — Lottery-based identification with verified balance
4. **Evidence on the counterfactual** — We do *not* observe the care our control group receives, but we characterise it using aggregate enrollment data and an independent survey of Barcelona families with young children that identifies rejected applicants

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Appendix: What Does 0.08 SD Mean?



Kraft (2020) benchmarks for education interventions:

- **Small:** <0.05 SD — most scaled-up interventions
- **Medium:** **0.05–0.20 SD** — class size reduction, tutoring
- **Large:** >0.20 SD — intensive early childhood programs (Perry, Abecedarian)

Appendix: English as a Placebo

Why no effect on English?

- Children under 3 in Barcelona are **not exposed to English**
- Neither at home nor in childcare¹
- English proficiency at age 12 is determined by formal schooling (age 3+)

If the counterfactual is (partly) home care:

- Home provides richer Catalan/Spanish exposure
- But no English advantage either way

The null on English supports the interpretation that effects operate through **early language exposure**

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¹Public nurseries used Catalan only (1983 Linguistic Normalization Act). Private bilingual English nurseries were rare and served mainly expat families. According to the *Enquesta d'Usos Lingüístics* (2008), 90% of Barcelona residents had Catalan, Spanish, or both as their first language.

Appendix: School Choice and School Fixed Effects

Admission to public childcare shifts later school enrollment:

- Increases attendance at public primary schools

But does school sorting explain our results? Controlling for school fixed effects:

	(1)	(2)	(3)	(4)	(5)
	Average Test Score	Test Score Catalan	Test Score Spanish	Test Score English	Test Score Maths
Admitted	-0.082*** (0.029)	-0.062** (0.028)	-0.080*** (0.029)	-0.019 (0.030)	-0.099*** (0.029)
Observations	7980	7980	7980	7980	7980
Mean	0.00	0.00	0.00	0.00	0.00

Results are **unchanged** $\Rightarrow$ Effects not driven by sorting into different schools

Peer quality $\rightarrow$

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Appendix: Strategic Applications and the Boston Mechanism

Barcelona uses a **Boston-style** mechanism: families may not report their true preferences.

Two forms of strategic behaviour [cf. Drange and Havnes, 2019]:

- Applying to centers with a **higher probability of an offer** (low oversubscription)
- Ranking such a center **first**, even if it is not the most preferred

Internal validity is unaffected: conditional on the risk set (center $\times$ year $\times$ level), the lottery number is random, so strategic ranking does not affect *who* wins.

It matters for interpretation and external validity.

Planned: heterogeneity by **ex-ante admission probability**, comparing lotteries at very high- vs very low-probability centers.

What Happens to Those Not Admitted?

Families not admitted to their first choice. What now?

- They may be admitted to a **lower-preference center** (but only $\sim 3\%$ are)
- More likely: they must find **alternative care**
 - Private childcare (expensive)
 - Parental care (mother, father)
 - Grandparents or other family
 - Nanny or informal care

This is the counterfactual we examine in the mechanism section.

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Aggregate and survey data (not linked at individual level):

- **Neighborhood characteristics (IDESCAT)**
 - Socioeconomic indicators at *agrupació censal* level
 - 190 agrupacions censals in Barcelona
- **District-level enrollment (Generalitat de Catalunya)**
 - Public/private childcare enrollment by district (10 districts)
 - Combined with newborn data to compute enrollment shares
- **Childcare arrangements survey (IGOP, 2020)**
 - 1,219 families with young children in Barcelona
 - Identifies those who applied to public nurseries and were rejected, and the care they used instead

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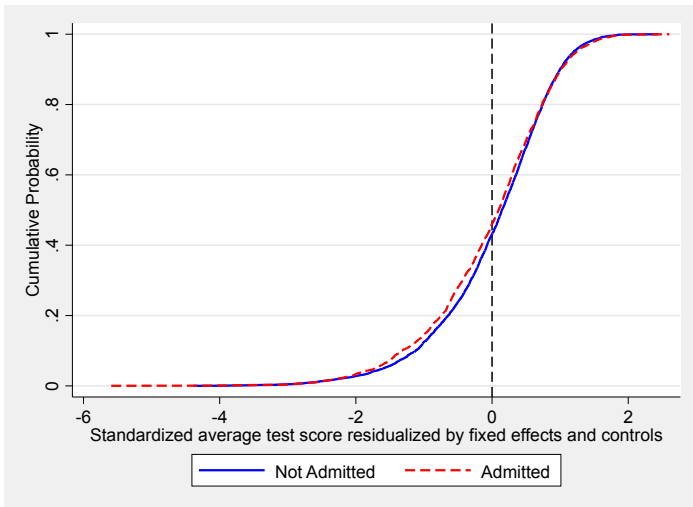
Sample Construction

Sample restrictions applied to the IDESCAT-linked individual-level data:

Restriction	% dropped	Obs. remaining
Full sample	—	23,817
(1) No missing pre-treatment variables	12.6%	20,818
(2) No missing outcome variables	17.9%	16,557
(3) Valid application	1.2%	16,260
(4) No special educational needs	0.2%	16,217
(5) No simultaneous sibling applications	5.8%	14,830
(6) Priority score = 30 (catchment area only)	14.3%	11,419
(7) First-choice school held lottery	14.4%	7,980

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Distribution of Effects



CDF for admitted lies **to the left** — first-order stochastic dominance.
Effect is systematic across the distribution.

Is Our Sample Also Selected?

Is our sample very affluent? Yes, somewhat — but far less extreme than Fort et al.

Variable	Mean Lottery	Mean Applicants	Difference	p-value
<i>Panel A: Individual and family characteristics</i>				
# Preferences expressed	2.39	2.03	0.30***	0.00
Female child	0.48	0.48	0.01	0.31
Age child at application	1.21	1.20	0.01	0.25
# siblings who applied before	0.06	0.12	-0.08***	0.00
Age Mother	33.74	33.73	0.20**	0.02
Age Father	35.74	36.00	-0.08	0.41
Spanish nationality Mother	0.83	0.73	0.14***	0.00
Spanish nationality Father	0.82	0.72	0.15***	0.00
Mother has Bachelor Degree	0.70	0.67	0.07***	0.00
Father has Bachelor Degree	0.63	0.63	0.04***	0.00
Both parents with a Bachelor Degree	0.53	0.52	0.05***	0.00
Resides with Mother	0.91	0.92	-0.02***	0.00
Resides with Father	0.85	0.85	-0.01	0.35
Resides with both parents	0.81	0.82	-0.02**	0.01
<i>Panel B: Neighborhood characteristics (agrupació censal)</i>				
Foreigners from Low-income countries (%)	12.50	12.77	-0.13	0.13
Population w/o post-secondary (%)	15.40	14.41	-0.06	0.24
Young w/o post-secondary (%)	21.62	20.88	-0.66***	0.00
Employed population (%)	64.98	64.78	0.06	0.27
Average income (€)	14,648.17	15,212.49	141.74***	0.00
Low qualified workers (%)	9.43	9.06	-0.31***	0.00
Observations	7,980	20,818		

Lottery sample has higher education, more Spanish nationals — since disadvantaged families get priority points and skip the lottery.

Who Applies? Survey Evidence

Comparing applicants to the general population:

	Mean Applicants	Mean Non-applicants	Difference	p-value
<i>Panel A: Mother characteristics</i>				
Mother has university degree	0.513	0.451	0.062**	0.030
Mother born abroad	0.120	0.149	-0.028	0.146
Mother born in Barcelona	0.456	0.392	0.064**	0.024
<i>Panel B: Labor and income characteristics</i>				
Mother worked before birth	0.929	0.910	0.019	0.232
Mother was employee	0.463	0.353	0.110	0.290
Had permanent contract	0.526	0.444	0.082	0.630
Income > 25,000 EUR	0.554	0.476	0.078***	0.008
Mother is main earner	0.542	0.587	-0.045	0.122
<i>Panel C: Household and partner characteristics</i>				
Number of children (0-18)	1.63	1.56	0.07*	0.092
Children born since 2016	1.18	1.14	0.04*	0.089
Partner has university degree	0.463	0.387	0.077***	0.008
Child's birth year	2017.01	2017.37	-0.36***	0.000
Observations	616	599		

Applicants are positively selected — but still a **broad, heterogeneous pool**, not an elite slice.

Appendix: Characteristics of Public Childcare

Fees (during study period, 2005–2010):

- Highly subsidized: families paid ~**240€/month** (flat fee)²
- Discounts of 30%, 50%, or 80% for low-income families
- Includes lunch and afternoon snack
- Private alternative: 400–600€/month

Schedule:

- Full-day: 9:00–17:00 (8 hours)
- Part-time options available (minimum 4 hours/day)
- Extended hours (until 18:00) available through parent associations

Calendar:

- Academic year: September to mid-July (~10.5 months)
- Follows standard school calendar (Christmas, Easter breaks)
- Children admitted from age 4 months (16 weeks)

Appendix: Test Score Assessment

Overview:

- Standardized assessment at end of primary education (6th grade, age 12)
- Census-based: **all students** in Catalonia take it (not a sample)
- External evaluation: graded by teachers from other schools
- Mandatory but has no effect on student grades

Subjects tested:

- Catalan language (*competència lingüística en llengua catalana*)
- Spanish language (*competència lingüística en llengua castellana*)
- English language (*competència lingüística en llengua anglesa*)
— added 2010
- Mathematics (*competència matemàtica*)

History: First administered in 2009. Data available from 2009–present.

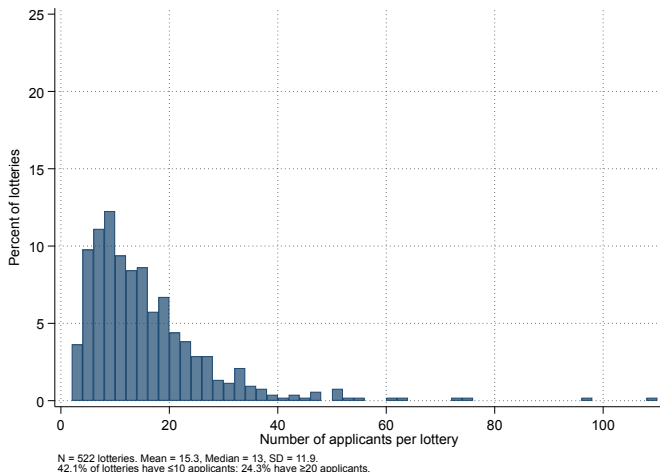
Appendix: Distribution of Priority Points

Share of applicants with each priority criterion:

	%
Home in Catchement area	92.93
Home in Municipality	2.24
Workplace in Catchement area	1.97
Siblings in Childcare center	5.00
Guaranteed Minimum Income Beneficiary	1.63
Disability	2.77
Large/single-parent families	10.77
Chronic Illness	1.10
<i>Exactly 30 points (catchment area only)</i>	75.60
Observations	60,648

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Appendix: Lottery Size Distribution



540 lotteries with median of 12 applicants per lottery.
57% of lotteries have more than 10 applicants; only 14% have 5 or fewer.

Appendix: Intent-to-Treat Interpretation

We estimate **ITT** — the effect of being *offered* a seat

- Not all admitted enroll ($\sim 90\%$ take-up)
- Some rejected get lower-preference slots ($\sim 3\%$)

Why ITT is policy-relevant:

- Effect of **expanding capacity**
- When you add seats, who benefits/loses at the margin?
- The 30-point applicants are exactly those who would gain from expansion

[First choice](#) → [Results with T_all](#) →

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Appendix: Assignment to First Choice

What share of lottery participants get assigned to their first choice?

Assignment outcome	N	%
Assigned to 1st preference	2704	33.9
Assigned to lower preference		
2nd preference	181	2.3
3rd preference	44	0.6
4th preference	12	0.2
5th preference	4	0.1
Not assigned to any school	5035	63.1
Total	7980	100.0

34% assigned to first choice; 63% not assigned anywhere.

Only 3% assigned to lower preferences $\Rightarrow$ minimal “contamination” of control group.

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Appendix: Balancing Test

Variable	Mean Admitted	Mean Rejected	Difference	p-value
<i>Panel A: Individual and family characteristics</i>				
# Preferences expressed	2.45	2.36	-0.01	0.67
Female child	0.48	0.48	-0.02	0.23
Age child at application	1.29	1.18	-0.01	0.41
# siblings who applied before	0.06	0.06	0.00	0.62
Age Mother	33.60	33.81	0.14	0.24
Age Father	35.60	35.82	0.09	0.54
Spanish nationality Mother	0.83	0.83	0.00	0.87
Spanish nationality Father	0.83	0.82	-0.00	0.68
Mother has Bachelor Degree	0.66	0.72	0.00	0.72
Father has Bachelor Degree	0.59	0.65	-0.01	0.33
Both parents with a Bachelor Degree	0.48	0.55	0.00	0.85
Resides with Mother	0.92	0.91	0.00	0.86
Resides with Father	0.85	0.86	-0.01	0.13
Resides with both parents	0.81	0.81	-0.01	0.18
<i>Panel B: Neighborhood characteristics (agrupació censal)</i>				
Foreigners from Low-income countries (%)	13.14	12.17	0.00	0.99
Population w/o post-secondary (%)	16.47	14.86	-0.08	0.28
Young w/o post-secondary (%)	23.96	20.42	-0.06	0.71
Employed population (%)	64.42	65.26	0.05	0.51
Average income (€)	13,988.66	14,986.18	45.10	0.38
Low qualified workers (%)	10.55	8.85	-0.06	0.47
Observations	2,704	5,276		

No statistically significant differences. Randomization successful.

Appendix: Attrition Check

Are admitted and rejected children equally likely to appear in test score data?

Variable	Mean Admitted	Mean Rejected	Difference	p-value
Attrition (all levels)	0.18	0.20	-0.01	0.47
Age 0-1	0.17	0.18	-0.01	0.63
Age 1-2	0.17	0.20	-0.01	0.56
Age 2-3	0.19	0.22	-0.00	0.79
Observations	3307	6622		

No differential attrition. Results are not driven by selective sample loss.

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Appendix: Balancing Test – Level i0

Variable	Mean Admitted	Mean Rejected	Difference	p-value
<i>Panel A: Individual and family characteristics</i>				
# Preferences expressed	2.55	2.36	-0.08	0.47
Female child	0.51	0.48	0.01	0.77
Age child at application	0.30	0.30	-0.01	0.34
# siblings who applied before	0.02	0.03	-0.00	0.81
Age Mother	33.45	33.61	0.22	0.55
Age Father	35.07	35.29	-0.11	0.78
Spanish nationality Mother	0.85	0.88	-0.02	0.54
Spanish nationality Father	0.83	0.87	-0.03	0.32
Mother has Bachelor Degree	0.80	0.82	0.02	0.60
Father has Bachelor Degree	0.69	0.74	-0.00	0.90
Both parents with a Bachelor Degree	0.60	0.67	0.00	0.97
Resides with Mother	0.87	0.88	-0.01	0.77
Resides with Father	0.83	0.82	0.01	0.81
Resides with both parents	0.77	0.76	0.00	0.93
<i>Panel B: Neighborhood characteristics (agrupació censal)</i>				
Foreigners from Low-income countries (%)	12.89	11.35	0.38	0.12
Population w/o post-secondary(%)	15.70	13.45	0.27	0.18
Young w/o post-secondary (%)	21.56	17.28	0.77*	0.06
Employed population(%)	64.57	65.90	-0.23	0.19
Average income (€)	14,438.80	15,691.29	-187.35	0.15
Low qualified workers (%)	9.43	7.30	0.17	0.37
Observations	265	528		

Balance for applicants at level i0 (ages 0–1)

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Appendix: Balancing Test – Level i1

Variable	Mean Admitted	Mean Rejected	Difference	p-value
<i>Panel A: Individual and family characteristics</i>				
# Preferences expressed	2.59	2.46	0.05	0.33
Female child	0.47	0.47	-0.02	0.32
Age child at application	0.93	0.93	-0.00	0.82
# siblings who applied before	0.04	0.05	0.02*	0.06
Age Mother	33.37	33.75	0.08	0.64
Age Father	35.46	35.74	0.19	0.31
Spanish nationality Mother	0.87	0.84	0.00	0.74
Spanish nationality Father	0.86	0.83	0.01	0.65
Mother has Bachelor Degree	0.71	0.73	0.03	0.13
Father has Bachelor Degree	0.61	0.67	-0.01	0.69
Both parents with a Bachelor Degree	0.51	0.56	0.01	0.69
Resides with Mother	0.92	0.91	0.01	0.36
Resides with Father	0.86	0.86	-0.01	0.45
Resides with both parents	0.82	0.82	-0.01	0.64
<i>Panel B: Neighborhood characteristics (agrupació censal)</i>				
Foreigners from Low-income countries (%)	13.35	12.43	0.04	0.83
Population w/o post-secondary(%)	16.60	15.13	-0.19*	0.09
Young w/o post-secondary (%)	24.70	20.89	-0.19	0.48
Employed population(%)	64.12	65.04	0.07	0.49
Average income (€)	13,791.89	14,849.20	47.67	0.49
Low qualified workers (%)	10.83	9.09	-0.08	0.50
Observations	1,257	3,083		

Balance for applicants at level i1 (ages 1–2)

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Appendix: Balancing Test – Level i2

Variable	Mean Admitted	Mean Rejected	Difference	p-value
<i>Panel A: Individual and family characteristics</i>				
# Preferences expressed	2.29	2.18	-0.08	0.10
Female child	0.49	0.51	-0.02	0.37
Age child at application	1.89	1.90	-0.01	0.40
# siblings who applied before	0.08	0.10	-0.01	0.46
Age Mother	33.88	34.01	0.19	0.32
Age Father	35.86	36.13	0.02	0.94
Spanish nationality Mother	0.80	0.78	0.00	0.86
Spanish nationality Father	0.80	0.78	-0.01	0.60
Mother has Bachelor Degree	0.59	0.66	-0.02	0.23
Father has Bachelor Degree	0.54	0.60	-0.02	0.31
Both parents with a Bachelor Degree	0.42	0.48	-0.00	0.88
Resides with Mother	0.92	0.93	-0.01	0.54
Resides with Father	0.84	0.86	-0.03*	0.08
Resides with both parents	0.81	0.82	-0.03*	0.09
<i>Panel B: Neighborhood characteristics (agrupació censal)</i>				
Foreigners from Low-income countries (%)	12.98	11.93	-0.14	0.41
Population w/o post-secondary(%)	16.50	14.78	-0.04	0.71
Young w/o post-secondary (%)	23.72	20.54	-0.13	0.62
Employed population(%)	64.71	65.49	0.08	0.42
Average income (€)	14,096.99	15,020.61	101.41	0.24
Low qualified workers (%)	10.51	8.88	-0.09	0.48
Observations	1,182	1,670		

Balance for applicants at level i2 (ages 2–3)

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Appendix: Balancing Test – Alternative Treatment (T_all)

Variable	Mean Admitted	Mean Rejected	Difference	p-value
<i>Panel A: Individual and family characteristics</i>				
# Preferences expressed	2.52	2.32	0.12***	0.00
Female child	0.48	0.48	-0.02	0.13
Age child at application	1.29	1.17	-0.01	0.26
# siblings who applied before	0.06	0.06	0.00	0.53
Age Mother	33.58	33.84	0.12	0.33
Age Father	35.53	35.86	0.02	0.86
Spanish nationality Mother	0.83	0.83	-0.00	0.96
Spanish nationality Father	0.83	0.82	-0.01	0.56
Mother has Bachelor Degree	0.67	0.72	0.01	0.32
Father has Bachelor Degree	0.59	0.66	-0.01	0.39
Both parents with a Bachelor Degree	0.48	0.55	0.01	0.46
Resides with Mother	0.92	0.91	0.00	0.80
Resides with Father	0.85	0.86	-0.01	0.17
Resides with both parents	0.81	0.81	-0.01	0.28
<i>Panel B: Neighborhood characteristics (agrupació censal)</i>				
Foreigners from Low-income countries (%)	13.21	12.08	-0.04	0.73
Population w/o post-secondary(%)	16.49	14.77	-0.10	0.16
Young w/o post-secondary (%)	23.95	20.26	-0.17	0.32
Employed population(%)	64.40	65.32	0.07	0.30
Average income (€)	13,965.37	15,047.54	57.68	0.24
Low qualified workers (%)	10.59	8.75	-0.08	0.33
Observations	2,945	5,035		

Balance using T_all: admitted to **any** listed center (not just first choice)

Appendix: Grade Retention

	(1)	(2)	(3)
	Grade Retainer	Grade Retainer Primary	Grade Retainer Secondary
Admitted	0.000 (0.004)	-0.002 (0.003)	0.002 (0.002)
Observations	7980	7980	7926
Mean	0.02	0.01	0.01

Point estimates are essentially zero and statistically insignificant. However, grade retention is rare (2%), limiting statistical power.

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Appendix: Main Results – Alternative Treatment (T_all)

	(1) Average Test Score	(2) Test Score Catalan	(3) Test Score Spanish	(4) Test Score English	(5) Test Score Maths
Admitted	-0.068*** (0.026)	-0.048* (0.026)	-0.074*** (0.027)	-0.016 (0.027)	-0.078*** (0.028)
Observations	7980	7980	7980	7980	7980
Mean	0.00	0.00	0.00	0.00	0.00

Results using T_all: admitted to **any** listed center (not just first choice)
Effects remain negative and significant

← ITT ← Results

Appendix: Literature Comparison

Study	Country	Ages	Outcome	Effect (SD)	Design
<i>Panel A: Childcare for ages 0–2</i>					
This paper	Spain	0–2	Test scores, age 12	−0.08**	Lottery
Almås et al. (2026)	Norway	1–2	Maths, grade 9 (age 14)	+0.10**	IV (expansion)
Fort et al. (2020)	Italy	0–2	IQ, age 8–14	−0.10**	RD
Drange & Havnes (2019)	Norway	1–2	Test scores, age 7	+0.14**	Lottery
Felfe & Lalive (2018)	Germany	0–2	Dev. skills, age 6	+0.05*	IV
Baker et al. (2008)	Canada	0–4	Motor/social dev., age 0–3	−0.11**	DiD
<i>Panel B: Preschool for ages 3+</i>					
Berlinski et al. (2008)	Argentina	3–5	Test scores, age 8–15	+0.08**	DiD
Felfe et al. (2015)	Germany	3–6	School readiness, age 6–10	+0.05*	RD
Havnes & Mogstad (2015)	Norway	3–6	Years of education	+0.06**	DiD
Cascio & Schanzenbach (2013)	USA	4–5	Test scores, age 9–11	+0.15**	DiD
Blanden et al. (2016)	England	3	Test scores, age 11	≈ +0.03	DiD

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Appendix: Heterogeneity by Gender

	(1) Average Test Score	(2) Average Test Score	(3) Average Test Score	(4) Average Test Score
Admitted x Female	-0.034 (0.034)	-0.159* (0.092)	-0.024 (0.049)	-0.045 (0.047)
Admitted x Male	-0.126*** (0.036)	-0.093 (0.105)	-0.112** (0.052)	-0.149*** (0.044)
Observations	7980	792	4532	4248
Mean	-0.00	0.20	0.03	-0.06
Sample	All levels	Age 0–1	Age 1–2	Age 2–3
p-value (H0: equal)	0.04	0.61	0.17	0.08

Pooled: significantly larger effect for boys ($p = 0.04$). By level: difference not significant.

Contrasts with Fort et al. (2020), who find losses concentrated on girls; consistent with Baker et al. (2015), who find long-run negative effects concentrated on boys.

Appendix: Heterogeneity by Neighborhood Income

	(1) Average Test Score	(2) Average Test Score	(3) Average Test Score	(4) Average Test Score
Admitted $\times$ High Income	-0.071* (0.037)	-0.108 (0.089)	-0.116* (0.064)	-0.061 (0.046)
Admitted $\times$ Low Income	-0.092** (0.036)	-0.164 (0.120)	-0.045 (0.047)	-0.134*** (0.049)
Observations	7980	792	4532	4248
Mean	-0.00	0.20	0.03	-0.06
Sample	All levels	Age 0–1	Age 1–2	Age 2–3
p-value (H0: equal)	0.67	0.69	0.35	0.27

Negative effects in **both** high and low income neighborhoods.
Effects on low-income neighborhoods are slightly larger (-0.09 vs -0.08).

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Appendix: Heterogeneity by Month of Birth

	(1) Average Test Score	(2) Average Test Score	(3) Average Test Score	(4) Average Test Score
Admitted x Jan-June	-0.114*** (0.036)	-0.126* (0.074)	-0.071 (0.054)	-0.177*** (0.050)
Admitted x July-Dec	-0.052 (0.034)		-0.074 (0.046)	-0.044 (0.046)
Observations	7980	792	4532	4248
Mean	-0.00	0.20	0.03	-0.06
Sample	All levels	Age 0–1	Age 1–2	Age 2–3
p-value (H0: equal)	0.18	0.09	0.96	0.05

Children born Jan–June (older in cohort) show stronger negative effects (−0.11 SD).

Children born July–Dec (younger in cohort) show smaller, insignificant effects (−0.05 SD).

Difference is borderline significant for Age 2–3 ($p = 0.05$).

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Appendix: Population Enrollment Rates

What do children **in the general population** do at each age?

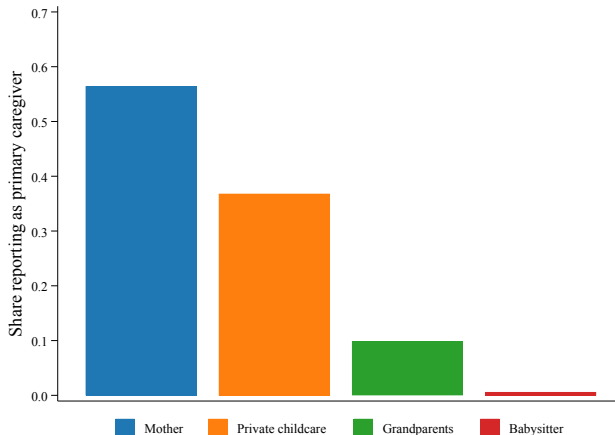
	Public	Private	No childcare
0–1	3.0%	7.8%	89.2%
1–2	9.5%	29.7%	60.8%
2–3	16.9%	45.7%	37.3%

At ages 0–1: 89% are in **no childcare** (parental/family care)

At ages 2–3: 45% in private, 17% in public, 37% no childcare

← Context ← Counterfactual

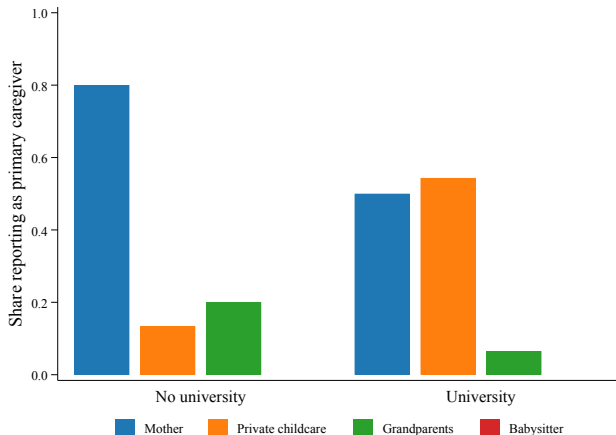
Appendix: Survey Evidence on Counterfactual Care



Among rejected families: mix of **parental care**, **private childcare**, and grandparents

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Appendix: Counterfactual by Mother's Education



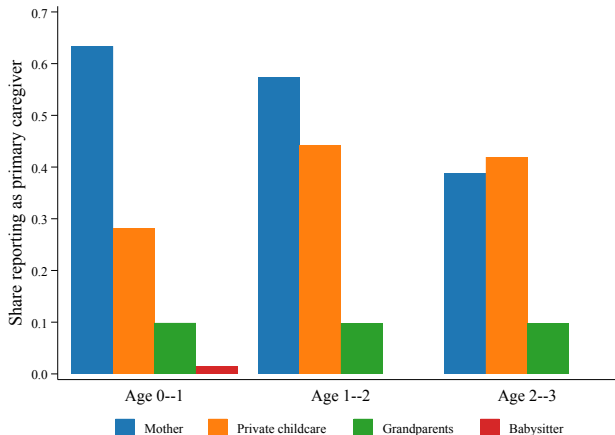
High-education mothers: more private childcare when rejected

Low-education mothers: more parental care when rejected

By child's age →

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Appendix: Counterfactual by Child's Age



Younger children: more parental care when rejected

Older children: more private childcare becomes available

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Appendix: Effect on Peer Test Scores

Do admitted children end up with lower-performing peers?

	(1) Average Test Score Peers	(2) Test Score Catalan Peers	(3) Test Score Spanish Peers	(4) Test Score English Peers	(5) Test Score Maths Peers
Admitted	-0.011 (0.009)	-0.002 (0.008)	-0.015* (0.008)	-0.007 (0.008)	-0.008 (0.009)
Observations	7801	7805	7804	7804	7807
Mean	0.02	0.01	0.01	0.02	0.02

No significant effect on average peer test scores in primary school.

⇒ Results are not driven by differences in school/peer quality.

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Appendix: Parental Education Variable

How is “High Education” defined?

Education is recorded using **CNES codes** (Catalan education classification):

- $\text{CNES} \geq 40$ = Post-secondary education (upper secondary, vocational, or university)
- $\text{CNES} < 40$ = No post-secondary education

Variable construction:

Variable	Definition
High Education	Both parents have $\text{CNES} \geq 40$
Low Education	At least one parent has $\text{CNES} < 40$

Distribution in lottery sample:

- Mother has $\text{CNES} \geq 40$: 66%
- Father has $\text{CNES} \geq 40$: 59%
- Both parents: 48%

Appendix: CNES Education Classification

Code	Description	Category
10-11	Cannot read or write	Low (<40)
20-21	No formal education / below primary	
22	Incomplete primary, 5 years EGB	
30	Primary complete or equivalent	
31	Lower secondary (EGB, ESO)	
32	Vocational training (first level)	
40	Upper vocational training (second level)	High (≥ 40)
41	Upper vocational, <i>mestria industrial</i>	
42	Upper secondary (<i>batxillerat</i> , BUP)	
43	Other intermediate qualifications	
44	<i>Diplomatura</i> (3-year university)	
45	Technical architect/engineer	
46	<i>Llicenciatura/Grau</i> (university degree)	
47	Non-university higher education	
48	Doctorate, Master's, postgraduate	
99	Unknown	Missing

“High Education” = Both parents have CNES ≥ 40

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Appendix: Private Availability Variable

How is “High Private Availability” defined?

Construction:

$$\text{Private Availability}_d = \frac{\text{Children in private childcare}_d}{\text{Children in public childcare}_d}$$

Classification: Median split at ratio = 2.20

District	Ratio	Classification
Sarrià-Sant Gervasi	4.85	High
Les Corts	3.42	High
Eixample	2.98	High
Gràcia	2.35	High
Sant Martí	2.21	High
Horta-Guinardó	1.89	Low
Sants-Montjuïc	1.76	Low
Sant Andreu	1.54	Low
Ciutat Vella	1.32	Low
Nou Barris	0.98	Low

Appendix: Results by Age Level (Including Re-Applicants)

	(1) Average Test Score	(2) Test Score Catalan	(3) Test Score Spanish	(4) Test Score English	(5) Test Score Maths
<i>Panel A: Age 0–1 (entry at 4–12 months)</i>					
Admitted	-0.126* (0.074)	-0.161** (0.076)	-0.175** (0.087)	-0.021 (0.077)	-0.041 (0.079)
Observations	792	792	792	792	792
Mean	0.20	0.10	0.08	0.18	0.24
<i>Panel B: Age 1–2 (entry at 12–24 months)</i>					
Admitted	-0.072* (0.039)	-0.080** (0.038)	-0.080** (0.040)	0.005 (0.037)	-0.076* (0.039)
Observations	4532	4532	4532	4532	4532
Mean	0.03	0.04	0.01	0.03	0.00
<i>Panel C: Age 2–3 (entry at 24–36 months)</i>					
Admitted	-0.098*** (0.034)	-0.037 (0.035)	-0.074** (0.033)	-0.076** (0.037)	-0.110*** (0.037)
Observations	4248	4248	4248	4248	4248
Mean	-0.06	-0.06	-0.02	-0.07	-0.05

Results are similar when including children who applied multiple times across years.

Appendix: Heterogeneity by Parental Education (Including Re-Applicants)

	(1) Average Test Score	(2) Average Test Score	(3) Average Test Score	(4) Average Test Score
Admitted x High Edu	-0.090*** (0.030)	-0.191** (0.074)	-0.076 (0.046)	-0.088* (0.045)
Admitted x Low Edu	-0.083** (0.036)	-0.014 (0.124)	-0.069 (0.059)	-0.107** (0.049)
Observations	9572	792	4532	4248
Mean	0.00	0.20	0.03	-0.06
Sample	All levels	Age 0–1	Age 1–2	Age 2–3
p-value (H0: equal)	0.87	0.17	0.92	0.76

Negative effects for both high- and low-education families persist with re-applicants included.

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